Math 571 - Homework 5

Richard Ketchersid

Problem 5.1 (R:5.8)**.** Suppose f' is continuous on $[a, b]$ and $\varepsilon > 0$. Show that there is $\delta > 0$ so that for all t such that $0 < |t - x| < \delta$ and all $a \leq x \leq b$

$$\left|\frac{f(t) - f(x)}{t - x} - f'(x)\right| < \varepsilon$$

This could be stated as f' is *uniform continuity* on $[a, b]$ provided f' is continuous on $[a, b]$. Does this hold for vector-valued functions?

f' is continuous on $[a, b]$ and hence uniformly continuous there since $[a, b]$ is compact. Fix $\varepsilon > 0$ and $\delta > 0$ so that $|f'(x) - f'(x')| < \varepsilon$ whenever $|x - x'| < \delta$. Let $t \in N_\delta(x)$, then MVT gives $c \in N_\delta(x)$ so that

$$\left|\frac{f(t) - f(x)}{t - x} - f'(x)\right| = |f'(c) - f'(x)| < \varepsilon$$

It $f : [a, b] \to \mathbb{R}^n$ (or $\mathbb{C}^n$) then this is still true as all of the component functions satisfy the conclusion. That is $f(x) = (f_1(x), \ldots, f_n(x))$ in the real case and $f_i(x) = u_i(x) + iv_i(x)$ in the complex case.

Problem 5.2 (R:5.16)**.** Suppose f is twice differentiable on $(0, \infty)$ and f'' is bounded on $(0, \infty)$, and $f(x) \to 0$ as $x \to \infty$. Show that $f'(x) \to 0$ as $x \to \infty$.

We have $f(x) = f(a) + f'(a)(x - a) + \frac{f''(c)}{2}(x - a)^2$ for some c between x and a. So $f'(a) = \frac{f(x)-f(a)}{x-a} - \frac{f''(c)}{2}(x - a)$. Let $x = a + h$ so we get

$$|f'(a)| \leq \left|\frac{f(a + h) - f(a)}{h}\right| + M|h|$$

Pick $\varepsilon > 0$. Fixing h we can make $Mh < \varepsilon/2$ and letting $a \to \infty$ we can make $|f(a + h)|, |f(a)| < h\varepsilon/4$ and thus

$$|f'(a)| \leq \frac{|f(a + h)|}{h} + \frac{|f(a)|}{h} + Mh < \varepsilon/4 + \varepsilon/4 + \varepsilon/2 = \varepsilon$$

Problem 5.3 (R:5.22)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. A point x is a **fixed point** of f iff $f(x) = x$.

(a) Show that if $f'(t) \neq 1$ for all $t \in (a, b)$, then f can have at most one fixed point.

If there were $x, y \in \mathbb{R}$ such that $x \neq y$, $f(x) = x$, and $f(y) = y$, then from MVT, there is c between x and y so that $f(x) - f(y) = x - y = f'(c)(x - y)$, but then $f'(c) = 1$.

(b) Show that $f(t) = t + (1+e^t)^{-1}$ satisfies $|f'(t)| < 1$ and f has no fixed points.

$f'(t) = 1 - \frac{e^t}{(1+e^t)^2}$, but $0 < \frac{e^t}{(1+e^t)^2} < 1$ so $0 < f'(t) < 1$.

It can't be the case that $f(t) = t$, since $t = t + (1+e^t)^{-1}$ would imply $(1+e^t)^{-1} = 0$ which is false.

(c) Show that if there is $A < 1$ so that $|f'(t)| \le A$ for all $t \in \mathbb{R}$, then f has a fixed point and moreover given any $x_0 \in \mathbb{R}$ and taking $x_{n+1} = f(x_n)$ it turns out that $x_n \to x$ and $f(x) = x$ is the unique fixed point of f.

$$|x_n - x_{n-1}| = |f(x_{n-1}) - f(x_{n-2})| = |f'(c)||x_{n-1} - x_{n-2}| \le A|x_{n-1} - x_{n-2}|$$

Continuing this gives

$$|x_n - x_{n-1}| \le A^{n-1}|x_1 - x_0|$$

and thus for $n > m$

$$|x_n - x_m| \le |x_n - x_{n-1}| + \cdots + |x_{m+1} - x_m| \le (A^{n-2} + \cdots + A^m)|x_1 - x_0|$$

Now $A^{n-1} + \cdots + A^m = A^m(A^{n-m-1} + \cdots + 1) = A^m\left(\frac{1-A^{n-m}}{1-A}\right) < A^m/(1-A)$ So for $\varepsilon > 0$, if N is chosen so that $A^N/(1-A) < \varepsilon$ and $m, n \ge N$, then

$$|x_n - x_m| < A^N/(1-A) < \varepsilon$$

Thus (x_n) is a C-seq and so $\lim_{n\to\infty} x_n = x$ exists and by continuity of f, $\lim_{n\to\infty} f(x_n) = f(x)$, but by definition $\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = x$ and thus $f(x) = x$. Uniqueness follows from (a).

(d) Show that the process described in (c) can be visualized by the zig-zag path

$$(x_1,\, x_2) \to (x_2,\, x_2) \to (x_2,\, x_3) \to (x_3,\, x_3) \to (x_3,\, x_4) \to \cdots.$$

Desmos visualization.

Problem 5.4 (R:5.24)**.** The process described in part (c) of Exercise 22 can of course also be applied to functions that map $(0, \infty)$ to $(0, \infty)$. Fix some $\alpha > 1$, and put

$$f(x) = \frac{1}{2}\left(x + \frac{\alpha}{x}\right), \qquad g(x) = \frac{\alpha + x}{1 + x}.$$

Both f and g have $\sqrt{\alpha}$ as their only fixed point in $(0, \infty)$. Try to explain, based on the basis of properties of f and g, why the convergence in Exercise 16, Chap. 3, is so much more rapid than it is in Exercise 17. (Compare f' and g', draw the zig-zags suggested in Exercise 22.) Do the same when $0 < \alpha < 1$.

Hint: A better hint than that given is to investigate using Taylor's Theorem with remainder with just three terms: $f'(x) = f(\sqrt{\alpha}) + f'(\sqrt{\alpha}))(x - \sqrt{\alpha}) + \frac{1}{2}f''(c)(x - \sqrt{\alpha})$ for some c between x and $\sqrt{\alpha}$.

There is no real change between what happens in the $0 < \alpha < 1$ and the $1 < \alpha$ cases. The key here is to look at what Taylor's Theorem tells us. Since $f'(\sqrt{\alpha}) = 0$ we have

$$f(x) - f(\sqrt{\alpha}) = f(\sqrt{\alpha}) + f'(\sqrt{\alpha})(x - \sqrt{\alpha}) + \frac{1}{2}f''(c)(x - \sqrt{\alpha})^2 = \frac{1}{2}f''(c)(x - \sqrt{\alpha})^2.$$

It is clear that $f''(c) = \frac{\alpha}{c^3}$ where $\alpha < c^2$ assuming that $x_0^2 > \alpha$. From this, we see $|x_{i+1} - \sqrt{\alpha}| < |x_i - \sqrt{\alpha}|^2$ and thus the number of correct digits doubles on each iteration.

Now look at what happens with g

$$g(x) - g(\sqrt{\alpha}) = g(\sqrt{\alpha}) + g'(\sqrt{\alpha})(x - \sqrt{\alpha}) + \frac{1}{2}g''(c)(x - \sqrt{\alpha})^2 = \frac{1 - \alpha}{(1 + \sqrt{\alpha})^2}(x - \sqrt{\alpha}) + \frac{\alpha - 1}{(1 + c)^3}(x - \sqrt{\alpha})^2.$$

Since $1 < \min\{\alpha, c\}$ and hence $g''(c) > 0$ we have $|x_{i+1} - \sqrt{\alpha}| < C|x_i - \sqrt{\alpha}|$ where $C = \frac{\alpha - 1}{(1+\sqrt{\alpha})^2}$. This is just some constant, for example, for $\alpha = 2$, $C = \frac{1}{(1+\sqrt{2})^2} \in (\frac{1}{9}, \frac{1}{8})$. So you gain a bit less than one decimal place per iteration.

Problem 5.5 (B5.2:2 - A fake counterexample to Theorem 5.9.)**.** The conclusion of Theorem 5.9 says, roughly, that if we look at the parametrized curve $(f(t), g(t))$, then for some value of t in (a, b), the ratio of the x- and y-components of its velocity will coincide with the ratio of the x- and y-components of the total displacement,$(f(b) - f(a), g(b) - g(a))$; i.e., that there is a point where the direction of the curve is parallel to the line segment from $(f(a), g(a))$ to $(f(b), g(b))$.

However, consider the case $f(t) = t^2$, $g(t) = t^3$ on the interval $[-1, 1]$. Sketch the graph, being especially careful near $t = 0$.

(a) Is there a point where the curve is parallel to the line segment referred to? For what value of t does the equation of Theorem 5.9 hold?

(b) How does the curve look at the corresponding point? (That is why I wrote "roughly" in the first sentence of this exercise.)

(c) Under what ***additional condition*** on the curve $r(t) = (f(t), g(t))$ for $t \in [a, b]$ can we remove the word roughly and exactly that at some $t_0 \in (a, b)$, $r'(t_0) = (f'(t_0), g'(t_0))$ is parallel to $(f(b) - f(a), g(b) - g(a))$?

Problem 5.6 (R:5.26)**.** Suppose $f(x)$ is differentiable on $[a, b]$, $f(a) = 0$, and there is a fixed A such that $|f'(x)| \leq A|f(x)|$ for all x in $[a, b]$. Show that $f(x) = 0$ on $[a, b]$.

Let $d = \sup\{x \in [a, b] \mid f|_{[a,x]} = 0\}$, by continuity it is clear that $f|_{[a,d]} = 0$. If $d = b$, we are done. If $d < b$ take $0 < \delta$ so that $\delta A < 1$ and $d + \delta \leq b$. Take $e \in (d, d + \delta]$. By MVT we have $\frac{f(e) - f(d)}{e - d} = f'(\hat{d})$ so that $|f(e)| = |f'(\hat{d})||e - d| \leq |f'(\hat{d})|\delta$.

Let $M = \sup(f[d, d + \delta])$ and $M' = \sup(f'[d, d + \delta])$ we know $M' \leq AM$ by assumption. On the other hand, we have just shown that $M \leq M'\delta$, so that $M \leq M'\delta < A\delta M < M$. A contradiction!

The next problem makes use of the one you just did.

Problem 5.7 (R:5:27)**.** Let $\phi : [a, b] \times [\alpha, \beta] \to \mathbb{R}$. A *solution to the initial–value problem* (IVP)

$$y' = \phi(x, y), \quad y(a) = c \text{ for } a \le c \le b$$

is a function $f : [a, b] \to [\alpha, \beta]$ satisfying

$$f(a) = c, \quad f'(x) = \phi(x, f(x)) \text{ for all } a \le x \le b$$

Show that if there is a constant $A \ge 0$ so that

$$|\phi(x, y_1) - \phi(x, y_2)| \le A|y_1 - y_2| \text{ for all } x \in [a, b] \text{ and } y_1, y_2 \in [\alpha, \beta],$$

then there is at most one solution to any such IVP.

Suppose f_1 and f_2 are two such solutions, then note that by assumption

$$|f_1'(x) - f_2'(x)| = |\phi(x, f_1(x)) - \phi(x, f_2(x))| \le A|f_1(x) - f_2(x)| \text{ for } x \in [a, b].$$

Letting $h(x) = f_1(x) - f_2(x)$ we have $h(a) = 0$, h is differentiable on $[a, b]$, and $|h'(x)| \le A|h(x)|$ for $x \in [a, b]$. Thus by Problem 1, $h = 0$ on $[a, b]$ and thus $f_1 = f_2$.

The book points out an example $y(0) = 0$ and $y' = y^{1/2}$ on $[0, 1]$. Note that this fails the hypotheses since there is no $A \ge 0$ with $|\sqrt{y}| < A|y|$ on $[0, 1]$, in particular, $\lim_{y \to 0^+} \frac{\sqrt{y}}{y} = \infty$. The book gives two solutions $y = 0$ and $y = x^2/4$. To find all solutions note

$$\begin{aligned}
\frac{y'}{y^{1/2}} &= 1 \\
y^{-1/2}\frac{dy}{dx} &= 1 \\
y^{-1/2}dy &= dx \\
\int y^{-1/2}\, dy &= \int dx \\
\frac{y^{1/2}}{1/2} + d &= x + c && (d \text{ and } c \text{ arbitrary constants}) \\
y^{1/2} &= \frac{x}{2} + C && (C \text{ an arbitrary constant}) \\
y &= \frac{x^2}{4} + 2Cx + C^2
\end{aligned}$$

If $y(0) = 0$, then $C^2 = 0$, so $C = 0$, and thus the two solutions are all.

Problem 5.8 (B:5.6:1 - *A function whose Taylor polynomials converge to the wrong result*)**.** Here is a well-known example concerning Taylor series, which Rudin doesn't give till Chapter 8 (where it is Exercise R:8.1), because it uses properties of the exponential function, which he develops there. Since we don't reach that chapter in this course, let us assume for this exercise a few basic properties of that function: That e^x is everywhere nonzero, that $e^{-x} = 1/e^x$, that e^x is differentiable, with derivative equal to itself, and that $e^x \to +\infty$ as $x \to +\infty$. We begin by deducing a few more facts from these.

(a) Show that for every polynomial $p(x)$, as $x \to +\infty$ we have $p(x)e^{-x} \to 0$. (Hint: Use L'Hospital's rule and induction on the degree of p.)

(b) Sketch an argument showing from this that for every polynomial $p(x)$, $\lim_{x\to 0} p(x^{-1})e^{-x^{-2}} = 0$. (I say "sketch" because, although Rudin proves in Theorem 4.7 that for continuous functions f and g with appropriate domains and codomains, the composite function $g \circ f$ is also continuous, he does not develop general results on *limits* of composite functions; in particular limits at infinity; and although it is not too difficult to prove such results, it would be time-consuming to have you do so here. So simply sketch how these functions behave, assuming that in this example, limits of composites behave as one would expect. Note that "$x \to 0$" involves values both above and below 0.) We can now give the example. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = e^{-x^2}$ if $x \neq 0$, and $f(0) = 0$.

(c) Show that on $\mathbb{R} \setminus \{0\}$, f has derivatives $f^{(n)}$ for all nonnegative integers n, and that for each n, $f^{(n)}(x)$ has the form $p_n(x^{-1})e^{-x^{-2}}$ for some polynomial $p_n(x)$.

(d) Show that for every nonnegative integer n, $f^{(n)}(0)$ is also defined, and equals 0. Now compute the Taylor polynomials shown in display (23) on p.110 for $a = 0$. Show that these converge as $n \to \infty$, but that the limit is not the original function $f(t)$.

(e) As a variant of the above, suppose we define a function g by $g(x) = e^{-x^{-2}}$ if $x > 0$, and $g(0) = 0$ if $x \leq 0$. Show that g is also infinitely differentiable, and is given by its Taylor series for all negative x, but not for any positive x.

Problem 5.9. For our finale, here are a couple of background things to look at Wikipedia and a Quanta article The Jagged, Monstrous Function That Broke Calculus.

Our final problem will build off of the last problem in Homework 4. Recall from there that $C([0,1]) = \{f : [0,1] \to \mathbb{R} \mid f \text{ is continuous}\}$ is a complete metric space under the ***uniform metric*** or ∞-norm. To review, $\|f\|_\infty = \sup\{f(x) \mid x \in [0,1]\}$ and by the extreme value theorem, this could be replaced with $\|f\|_\infty = \max f([0,1])$. We will use the Baire Category Theorem in $C([0,1])$ to show that a large set (co-meager set) of continuous functions are nowhere differentiable.

The goal here is not to look at a specific example of a continuous nowhere differentiable function, but rather to show that, in some sense, ***almost all*** continuous functions are nowhere differentiable.

Define

$$D = \{f \in C([0,1]) \mid f \text{ is differentiable at at least one point } x_0 \in [0,1]\}$$

Define

$$E_n = \{x \mid (\exists x \in [0,1])(\forall y \in [0,1])(|f(x) - f(y)| < n|x-y|\}$$

(a) Suppose $f \in C([0,1])$ is differentiable at a point $x_0 \in [0,1]$. Show that there is $n \in \mathbb{N}$ so that $|f(x) - f(x_0)| < n$ for all $x \in [0,1]$. That is $D \subseteq \bigcup_n E_n$.

(b) Show that E_n is closed. For this, take $f_i \in E_n$, suppose $f_i \to f$ uniformly. Show $f \in E_n$.

Hint: Take $x_i \in [0,1]$ so that $(\forall y \in [0,1])(|f_i(x_i) - f(y)| \leq n|x_i - y|)$. Since $[0,1]$ is compact find a subsequence x_{m_i} so that $x_{m_i} \to x \in [0,1]$. (sequential compactness).

(c) Show that $\text{int}(E_n) = \{\}$. That is, E_n is *nowhere dense*, or conversely, E_n^c is dense.

Hint: See this **visual hint (Desmos)**. Fix $\epsilon > 0$, find a piecewise linear $l(x) \in C([0,1])$ so that $\|f - l\|_\infty$ is small. Let $s(x)$ be a *sawtooth function*, on $[0,1]$ we have

$$s(x) = \begin{cases} 2x - \frac{1}{2} & \text{if } x \in [0, \frac{1}{2}] \\ -2x + \frac{3}{2} & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

Furthermore s is periodic, $s(x+1) = s(x)$. We can consider $as(bx)$ for b an integer with period $\frac{1}{b}$, or frequency b, and *amplitude* $\frac{a}{2}$. At any point $x_1 \in [0,1]$, $|as(bx) - as(bx_1)| = 2ab$ for so x with $|x - x_1| < \frac{1}{b}$. So by choosing a small, we get $\|as(bx)\|_\infty = \frac{|a|}{2}$ while at the same time by choosing b large we can ensure that $as(bx) \notin E_n$. Consider now $h(x) = l(x) + as(bx)$ as a point in $C([0,1])$ *close* to $f(x)$, but with the property that $|\frac{h(x)-h(y)}{x-y}|$ can be made arbitrarily large.

(d) Apply the Baire Category Theorem to see that the set of nowhere differentiable functions is a dense G_δ (***co-meager***) set in $C([0,1])$.

(a) Then fix δ so that $|f(x) - f(x_0)| < |x - x_0|$ for all $|x - x_0| < \delta$. Here we are just taking $\epsilon = 1$ and finding $\delta > 0$ so that $\frac{|f(x)-f(x_0)|}{|x-x_0|} < 1$ (weaker than being differentiable at x_0.) Since f is continuous on the compact set $[0,1]$ we know $|f(x)| < M$ for some M, $0 < M < \infty$. If $|x - x_0| \geq \delta$, then $|f(x) - x_0| < M = \frac{M}{\delta}\delta < \frac{M}{\delta}|x - x_0|$. So we see $|f(x) - f(x_0)| \leq \max\{1, \frac{M}{\delta}\} \leq n$ for some $n \in \mathbb{N}$.

(b) Take x_{m_i} as in the hint, then using continuity

$$|f(x) - f(y)| = \lim_i |f(x_{m_i}) - f(y)| \leq \lim_i n|x_{m_i} - y| = n|x - y|$$

So $f \in E_n$.

(c) See the visual provided above to get a feel for this argument. Take any $f \in E_n$. First, let $l(x)$ be a piecewise linear continuous approximation of f so that $|f(y) - l(y)| < \frac{\epsilon}{2}$ for all $y \in [0,1]$. Now consider the piecewise linear sawtooth $r(x) = \frac{\epsilon}{2}s(ax)$ and consider $h(x) = l(x) + s(x)$. It is clear that $|f(x) - h(x)| < \epsilon$ and that for any $x \in [0,1]$ for points y close to x, $|x - y| < \frac{1}{2a}$, $|h(x) - h(y)| \geq \frac{\epsilon}{2}(2a) + M = \epsilon a + M$ where M is the minimum of the absolute values of the slopes of the linear segments of l. By choosing a large, we can guarantee that $h \notin E_n$. This shows that for an $f \in E_n$ and any $\epsilon > 0$ we can find h with $\|f - h\|_\infty < \epsilon$ and $h \notin E_n$. So $f \notin \text{int}(E_n)$ and as f is arbitrary, $\text{int}(E_n) = \{\}$.

(d) We see that E_n^C is open dense for all n and since $C([0,1])$ is a complete metric space, BCT holds and $D^c = \bigcap E_n^c$ is dense. So the set of nowhere differentiable continuous functions is a dense G_δ.